WORKED SOLUTION

STEP III 1993, Question 15

Probability and Statistics • Waiting Time Distribution

STEP III 1993, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q15

The key idea

Focus on the *last* throw. The experiment stops at throw n exactly when that throw supplies the second of whichever face was still short, so the first $n-1$ throws must contain exactly one of that face and all the rest of the other. That makes the distribution a sum of two simple terms. The expectation is then a pair of series of the form $\sum n(n-1)x^{n-2}$, which we get by differentiating a geometric series twice.

Solution

We assume $0 < p < 1$ (otherwise the experiment never ends) and that throws are independent.

The distribution of N

We need at least two heads and two tails, so $N \geq 4$. Fix $n \geq 4$. On the event $N = n$, the nth throw is either a head or a tail, and these two cases are disjoint.

The nth throw is a head. Before it, the target was not reached, and throw n reached it by adding a head. So the first $n-1$ throws already contained at least two tails but only one head: exactly one head and $n-2$ tails (and $n-2 \geq 2$ because $n \geq 4$). Conversely, any such sequence followed by a head stops at exactly n. The single head can be in any of $n-1$ positions, so this case has probability

$$(n-1)\, p\, q^{n-2} \cdot p = (n-1)p^2q^{n-2}.$$

The nth throw is a tail. By the same argument with heads and tails swapped, the probability is $(n-1)q^2p^{n-2}$.

Adding,

$$\boxed{\mathrm{P}(N=n) = (n-1)(p^2q^{n-2} + q^2p^{n-2}), \qquad n = 4, 5, 6, \ldots}$$

and $\mathrm{P}(N=n) = 0$ for $n \leq 3$.

Ask yourself Why can't the first $n-1$ throws contain two heads when the last throw is a head?

If they did, they would contain two heads and (as the stop needs) two tails already, so the experiment would have ended before throw n. This is what pins the count at exactly one.

The expectation

For $|x|<1$, differentiating $\sum_{n\geq 0} x^n = (1-x)^{-1}$ twice gives

$$\sum_{n=2}^{\infty} n(n-1)x^{n-2} = \frac{2}{(1-x)^3}.$$

Our sums start at $n=4$, so we remove the $n=2$ and $n=3$ terms, which are 2 and $6x$:

$$\sum_{n=4}^{\infty} n(n-1)x^{n-2} = \frac{2}{(1-x)^3} - 2 - 6x.$$

Now

$$\mathrm{E}(N) = \sum_{n=4}^{\infty} n\,\mathrm{P}(N=n) = p^2 \sum_{n=4}^{\infty} n(n-1)q^{n-2} + q^2 \sum_{n=4}^{\infty} n(n-1)p^{n-2}.$$

Using the series with $x=q$ (so $1-x=p$) and then with $x=p$ (so $1-x=q$), both valid since $0<p,q<1$:

$$\begin{aligned}\mathrm{E}(N) &= p^2\Big(\frac{2}{p^3} - 2 - 6q\Big) + q^2\Big(\frac{2}{q^3} - 2 - 6p\Big)\\ &= \frac{2}{p} + \frac{2}{q} - 2(p^2+q^2) - 6pq(p+q).\end{aligned}$$

Since $p+q=1$ we have $\frac{1}{p}+\frac{1}{q} = \frac{p+q}{pq} = \frac{1}{pq}$ and $p^2+q^2 = (p+q)^2 - 2pq = 1-2pq$. Hence

$$\begin{aligned}\mathrm{E}(N) &= \frac{2}{pq} - 2(1-2pq) - 6pq\\ &= \frac{2}{pq} - 2 - 2pq = 2\Big(\frac{1}{pq} - 1 - pq\Big),\end{aligned}$$

as required.

Checks

- The probabilities sum to 1: the same series with one fewer factor gives $\sum_{n\geq 4}(n-1)p^2q^{n-2} = 1 - p^2 - 2p^2q$, and adding the swapped version leaves $1-(p^2+q^2)-2pq(p+q) = 1-1+2pq-2pq = 1$, using $p^2+q^2 = 1-2pq$.

- For a fair coin, $pq = \frac{1}{4}$ and $\mathrm{E}(N) = 2(4 - 1 - \frac{1}{4}) = \frac{11}{2}$. Directly: $\mathrm{P}(N = 4) = \frac{3}{8}$, $\mathrm{P}(N = 5) = \frac{1}{4}$, …, and the partial sums of $n\,\mathrm{P}(N = n)$ approach 5.5.
- As $p \to 0$, $\mathrm{E}(N) \approx 2/p$: we wait for two heads, each taking about $1/p$ throws.
- Enumerating all sequences up to length 10 reproduces the distribution exactly, and a simulation with $p = 0.3$ gave a mean of 7.10, against 7.1038 from the formula.

Common slips

- Writing $\binom{n}{2}$ instead of $n - 1$: the last throw's position is fixed, so only the lone head (or tail) among the first $n - 1$ throws is free.
- Forgetting to subtract the $n = 2$ and $n = 3$ terms when using the standard series.

Beyond STEP

N is a mixture of two shifted *negative binomial* waiting times, and the trick of differentiating a geometric series is a hand-made version of a *probability generating function*. With $G(s) = \mathrm{E}(s^N)$ you would get the mean as $G'(1)$ and the variance from $G''(1)$ just as quickly. Grimmett and Welsh, *Probability: An Introduction*, develops generating functions and waiting-time problems like this one.

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